

Advanced Econometric Methods:sample

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Question 1

1.1 Scatter plot

Figure 1 shows a positive association in simulated data. The line is the least-squares fit.

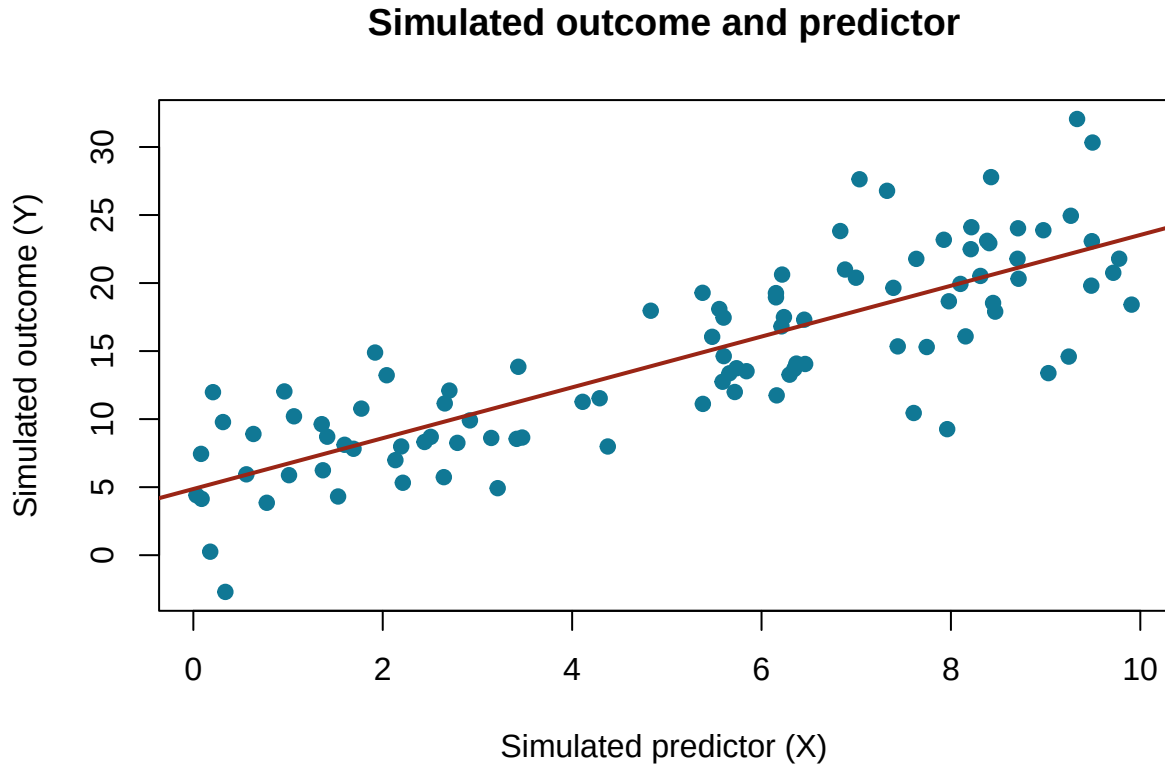


Figure 1: Simulated outcome (Y) and predictor (X), with a least-squares fit.

1.2 Event-study

The event-study simulation assigns treatment to 60 of 120 units. Outcomes include unit and time effects, a rising treatment effect after period 0, and random noise. Period -1 is the reference period. Figure 2 plots the estimated two-way fixed-effects coefficients and 95% confidence intervals; estimates before treatment should be near zero in this simulation.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k D_i \mathbb{1}\{t = k\} + \varepsilon_{it}.$$

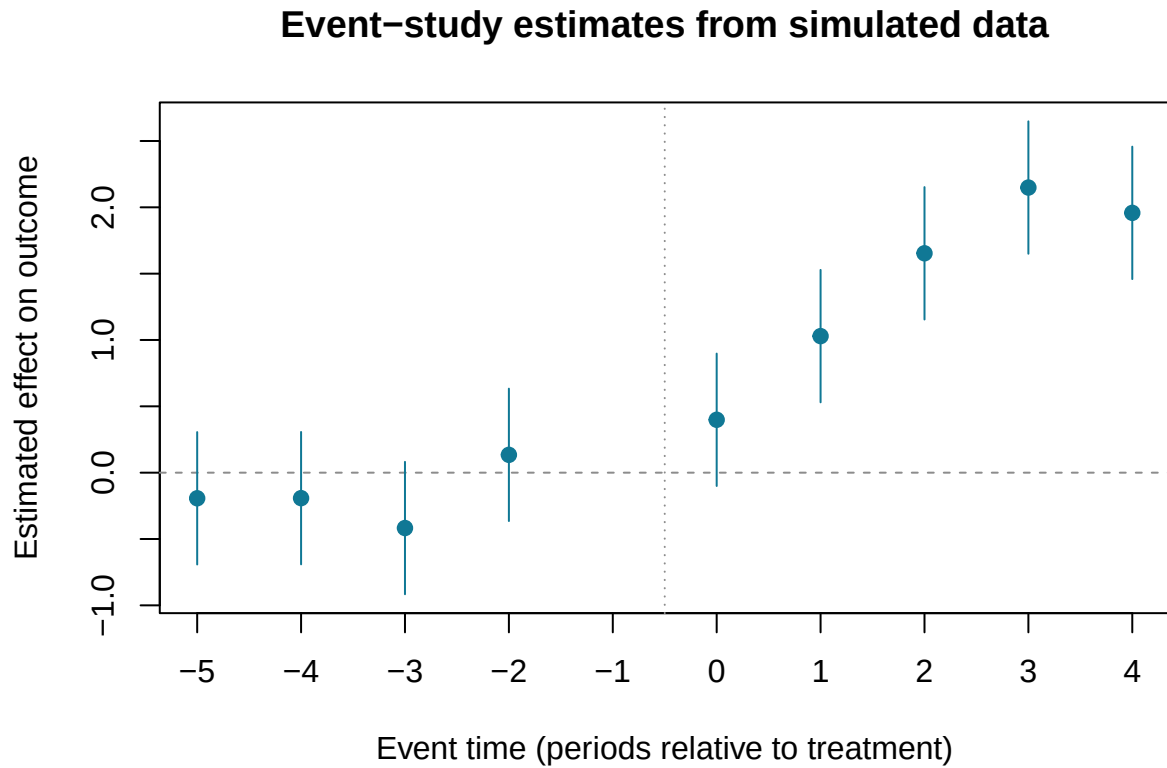


Figure 2: Event-study estimates from the simulated panel; bars show 95% confidence intervals.

Question 2

2.1 Penguin figure

Figure 3 uses the Palmer Penguins data. The panels show bill length by flipper length for each sex, body mass by flipper length, and the flipper-length distributions by species. Records with missing sex or plotted measurements are omitted.

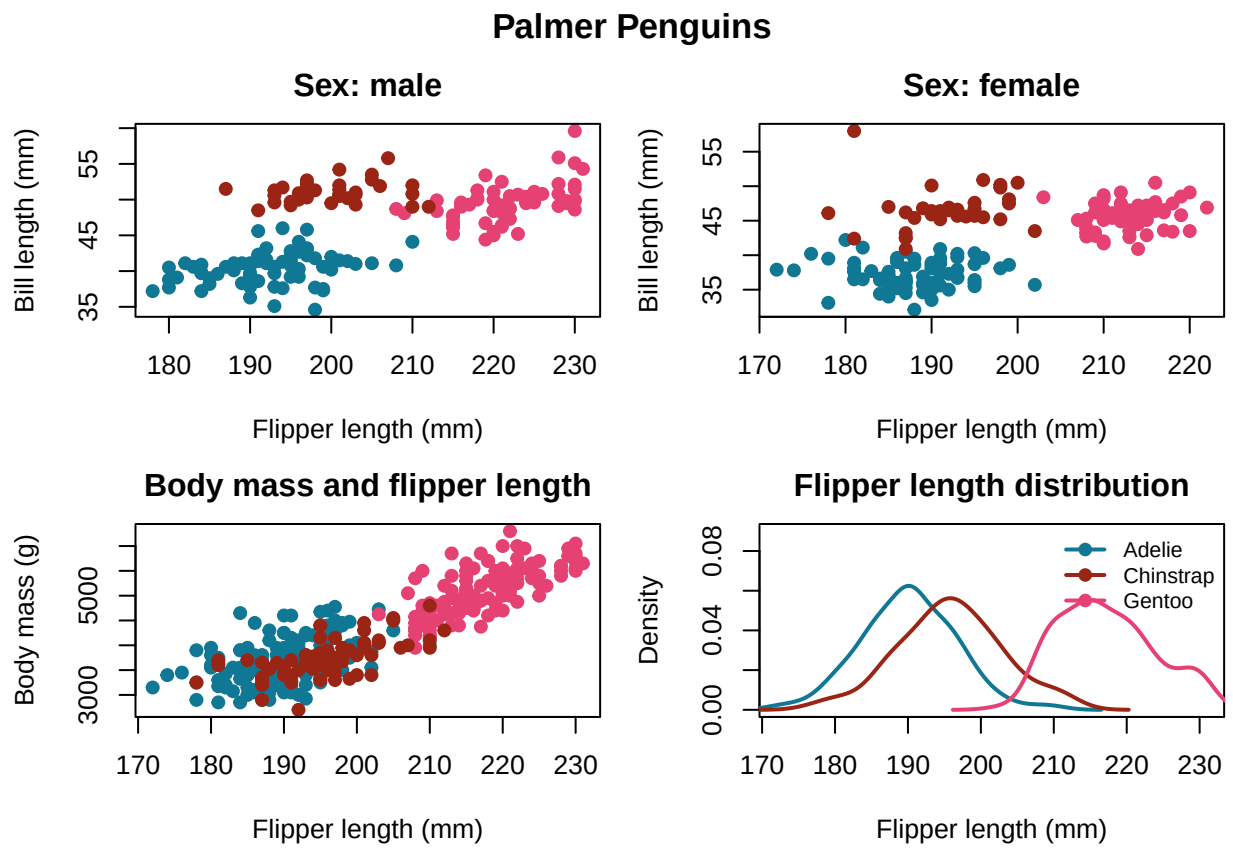


Figure 3: Penguin measurements by sex and species. Source: Horst, Hill, and Gorman (2020).

2.2 Table

Table 1 reports mean bill length, bill depth, flipper length, and body mass by species and sex. Records with missing sex are omitted; each mean uses the available observations for that measurement.

Table 1: Penguin summary statistics by species and sex.

	Species	Sex	Bill length (mm)	Bill depth (mm)	Flipper length (mm)	Body mass (g)
1	Adelie	Female	37.3	17.6	187.8	3368.8
4	Adelie	Male	40.4	19.1	192.4	4043.5
2	Chinstrap	Female	46.6	17.6	191.7	3527.2
5	Chinstrap	Male	51.1	19.3	199.9	3939.0
3	Gentoo	Female	45.6	14.2	212.7	4679.7
6	Gentoo	Male	49.5	15.7	221.5	5484.8

2.3 Regression

We estimate an OLS model of bill length on species, sex, and flipper length. The coefficients are conditional associations, not causal effects. Adelie and female penguins are the reference categories.

$$\text{Bill length}_i = \beta_0 + \beta_1 \mathbb{I}\{\text{Chinstrap}_i\} + \beta_2 \mathbb{I}\{\text{Gentoo}_i\} + \beta_3 \mathbb{I}\{\text{Male}_i\} + \beta_4 \text{Flipper length}_i + \varepsilon_i.$$

Table 2: OLS: bill length and penguin characteristics

	Bill length (mm)
Chinstrap vs. Adelie	9.434*** (0.354)
Gentoo vs. Adelie	5.974*** (0.650)
Male vs. female	3.004*** (0.289)
Flipper length (mm)	0.101*** (0.022)
Outcome mean	43.993
N	333
R ²	0.832
Outcome: bill length (mm). Complete cases only; omitted categories are Adelie and female. Conventional OLS standard errors in parentheses.	
• p < 0.1, ** p < 0.05, *** p < 0.01.	

The table reports conventional OLS standard errors in parentheses, the outcome mean, sample size, and R-squared. The sample contains complete cases for the variables in the regression.

Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). *palmerpenguins: Palmer Archipelago (Antarctica) penguin data* (R package version 0.1.0) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.3960218>